Capstone Weekly Achievement Report (CWAR)

Project Title: Smart POS Cafe Management System for Books and Blooms with Face Recognition Attendance

Reporting Period: September 13, 2026 – October 18, 2026

Week 1: September 13 - September 19, 2026

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1. Accomplishments

- Successfully conducted the initial client site visit at Books and Blooms to establish project baselines.
- Defined the core scope for the POS, Inventory, and Face Recognition Attendance modules.
- Initiated the drafting of the IEEE documentation framework.

2. Activities Performed

- Interviewed the client to understand the daily operational workflow of the cafe, specifically how books and floral products are tracked.
- Gathered specific requirements for the Face Recognition Attendance system for employees.
- Created the initial outline and Chapter 1 draft of the IEEE documentation.

3. Problems Encountered

- There was some uncertainty regarding how the client wanted to categorize bundled items (e.g., a book sold with a specific floral arrangement).

4. Next Plan

- Design the conceptual framework and database schema.
- Begin UI/UX wireframing for the POS interface.

Week 2: September 20 - September 26, 2026

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1. Accomplishments

- Finalized the relational database schema encompassing products, employees, and sales.
- Set up the development environment for the Face Recognition module.

2. Activities Performed

- Developed the backend database architecture.
- Began coding the basic POS transaction interface.
- Drafted Chapter 2 (Review of Related Literature) for the IEEE documentation.

3. Problems Encountered

- Initial testing of the face recognition library showed decreased accuracy under the specific warm, dim lighting conditions typical of the cafe environment.

4. Next Plan

- Develop the inventory management module.
- Collect more varied facial data under different lighting conditions to train and calibrate the attendance scanner.

Week 3: September 27 - October 3, 2026

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1. Accomplishments

- Successfully developed the initial prototype of the inventory system.
- Integrated basic stock deduction logic with the POS module.

2. Activities Performed

- Programmed the CRUD (Create, Read, Update, Delete) operations for the inventory backend.
- Connected the POS checkout process to the inventory database for real-time updates.
- Continued expanding the methodology section of the IEEE documentation.

3. Problems Encountered

- Encountered minor synchronization bugs where simultaneous simulated POS transactions caused race conditions in inventory stock deduction.

4. Next Plan

- Present the working inventory prototype to the client for evaluation.
- Debug and implement transaction locks to fix the inventory race conditions.

Week 4: October 4 - October 10, 2026

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1. Accomplishments

- Completed the client review session for the inventory system.
- Gathered actionable feedback and a list of new feature requests from the client.

2. Activities Performed

- Demonstrated the inventory and POS systems to the cafe owner.
- Documented the client's requests, which included adding specific category filters (Books vs. Blooms) and low-stock alerts.
- Refined the IEEE documentation to reflect the updated system architecture based on client input.

3. Problems Encountered

- The requested additional features (low-stock alerts and advanced filters) required structural modifications to the existing database tables.

4. Next Plan

- Restructure the database and implement the newly requested features.
- Optimize the Face Recognition attendance logging mechanism.

Week 5: October 11 - October 17, 2026

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1. Accomplishments

- Successfully fixed the needed features and integrated the client's requested additions into the system.
- Achieved stable performance for the Face Recognition Attendance module.

2. Activities Performed

- Implemented low-stock threshold alerts and advanced inventory filtering.
- Calibrated the face recognition system, adding a cooldown timer to prevent duplicate attendance logs.
- Drafted the system implementation and testing chapters of the IEEE documentation.

3. Problems Encountered

- The addition of real-time low-stock alerts caused a slight lag in the POS checkout rendering time.

4. Next Plan

- Optimize database queries to resolve the POS lag.
- Prepare the final build for the upcoming milestone presentation.

Week 6: October 18, 2026

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1. Accomplishments

- System performance optimized, resolving previous lag issues.
- IEEE documentation updated with all recent modifications and ready for review.

2. Activities Performed

- Conducted code refactoring and database query optimization for faster load times.
- Synchronized all system flowcharts and diagrams in the IEEE documentation with the finalized inventory and POS workflows.
- Performed end-to-end system testing encompassing POS, Inventory, and Attendance.

3. Problems Encountered

- Ensuring strict adherence to IEEE formatting guidelines for tables and figures required

manual adjustment.

4. Next Plan

- Conduct a final mock defense and system walk-through with the capstone adviser.
- Deploy the system for a pilot test at the actual cafe premises.